High Ion Load Laser Ion Source

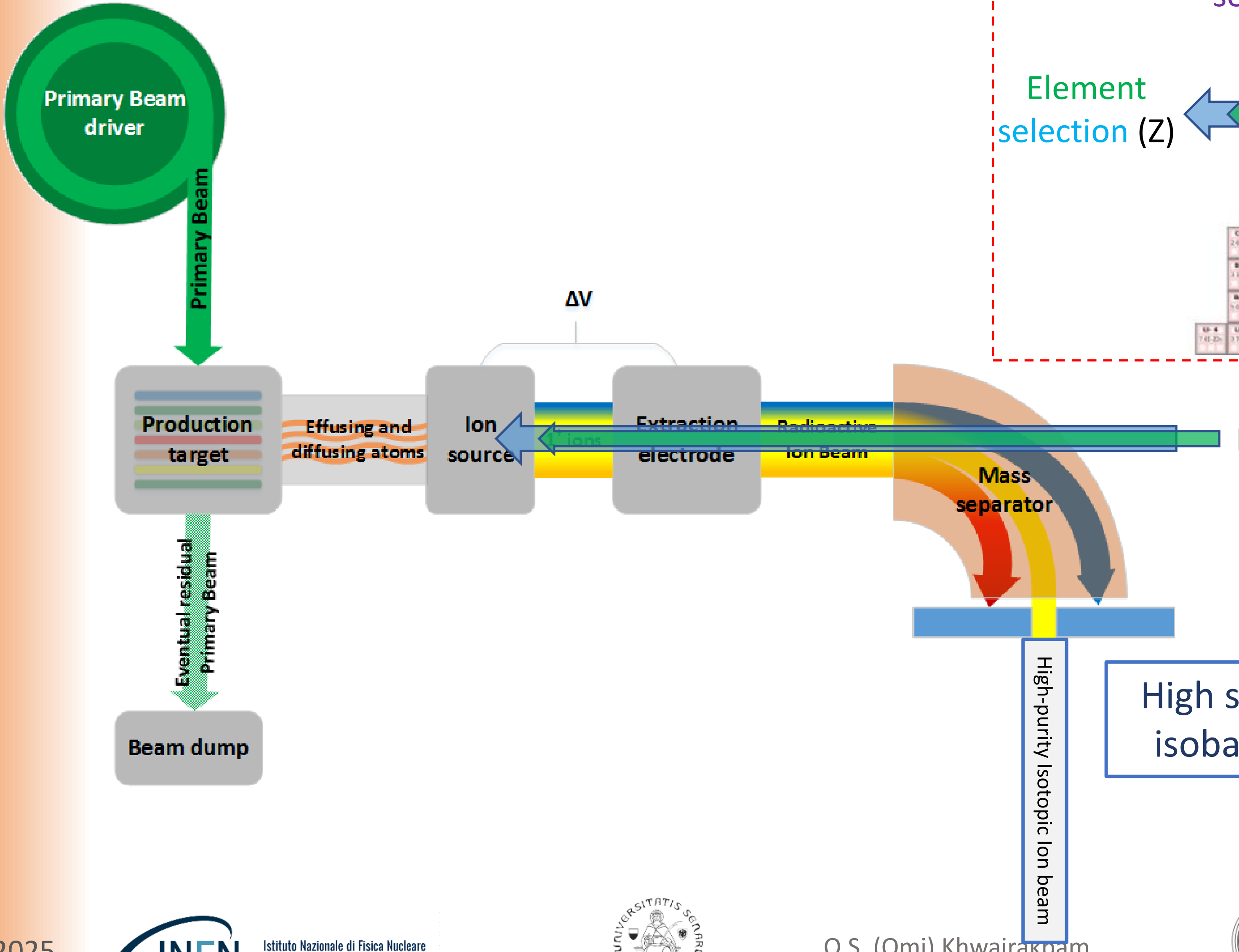
HILLIS

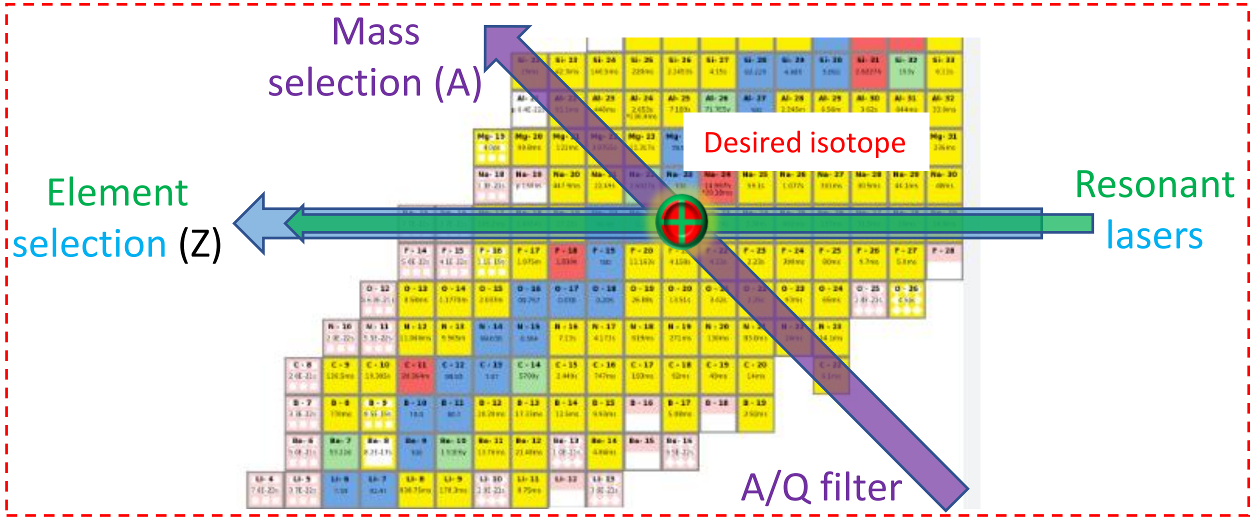
1st December, 2025

INFN Grant Giovani 2025

Dr. Omorjit Singh Khwairakpam (Omi)

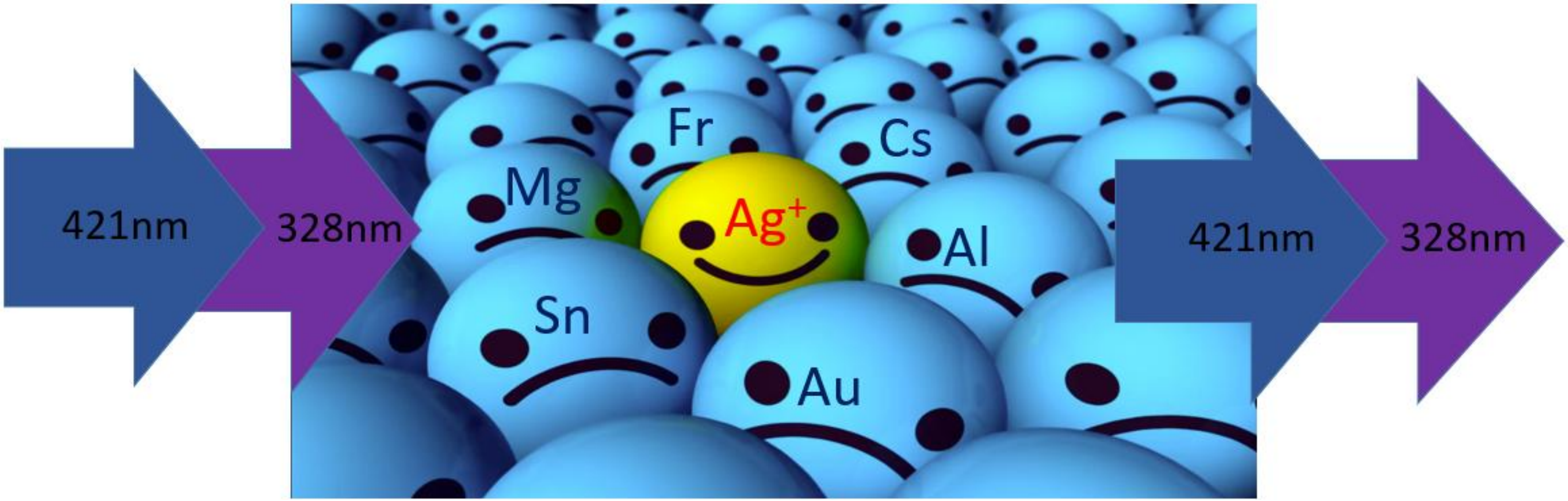
A simplified layout of an ISOL facility



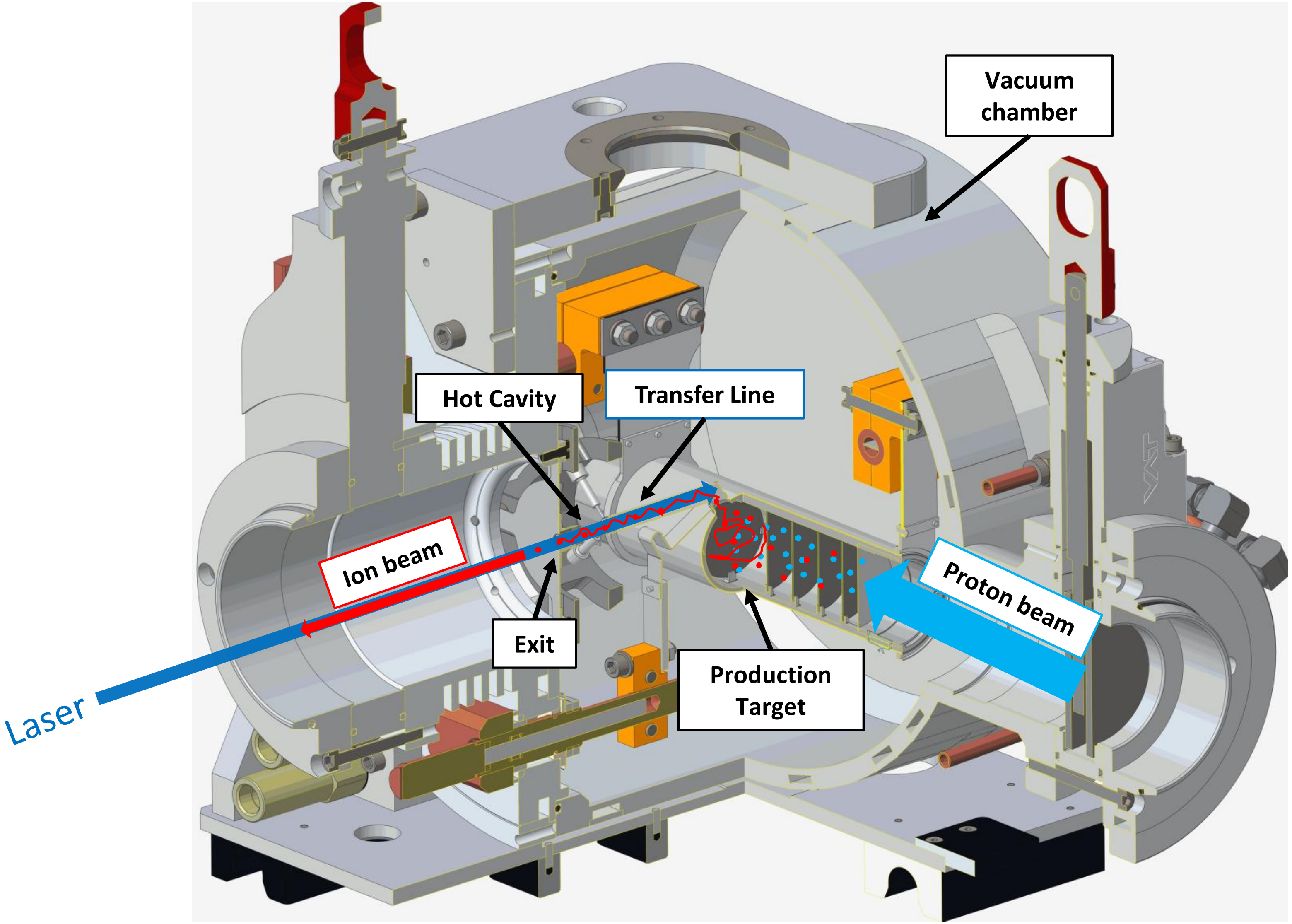


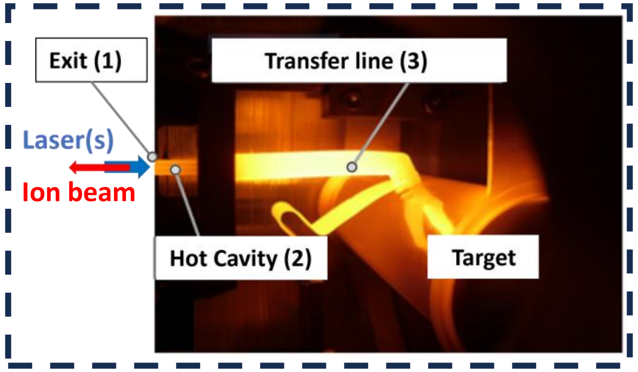
High suppression of the isobaric contaminants

Laser Photo-Ionization: In a nutshell

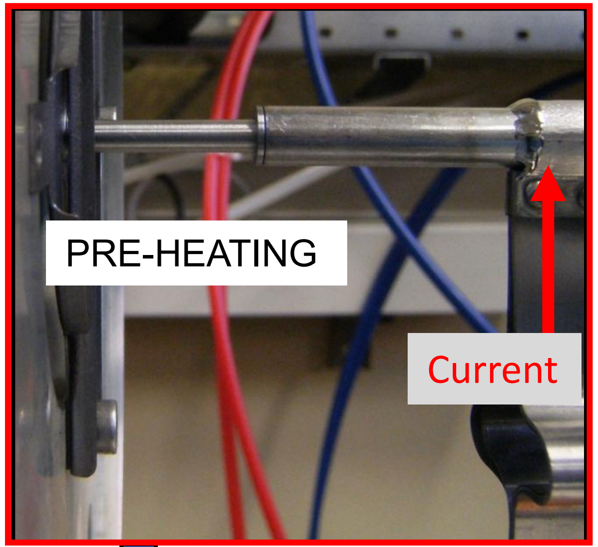


How does the Laser Ion Source work?

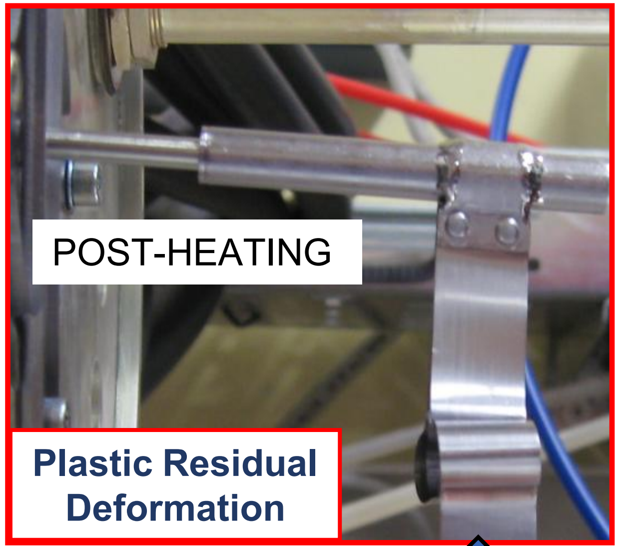


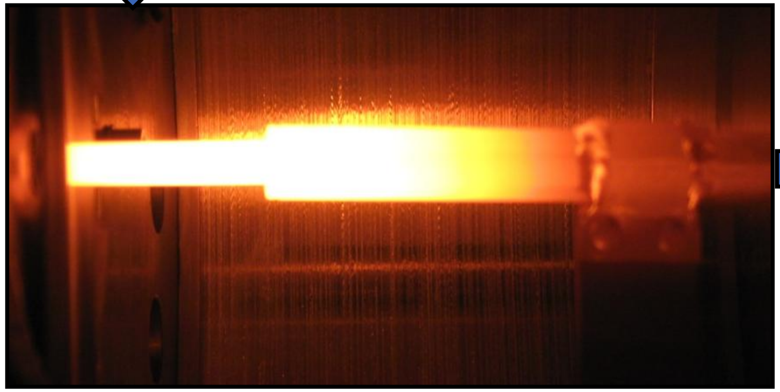


First SPES Hot Cavity Ion source

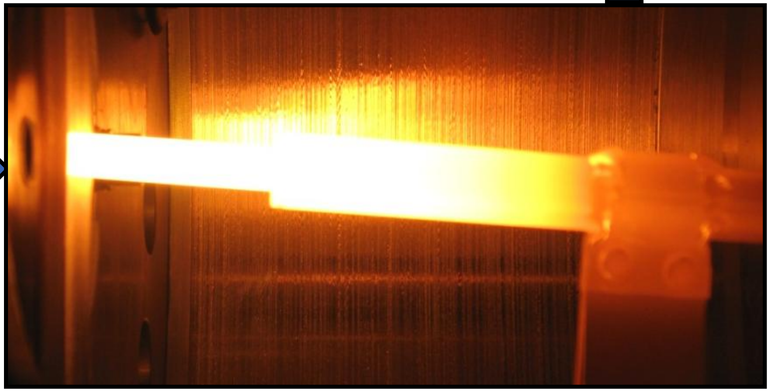


Adapted from ISOLDE MK1 source





DURING-HEATING (PRE-BENDING)



DURING-HEATING (POST-BENDING)

Wait…Why does the ion source need to be hot if the lasers do the ionization?

The answer lies in another question:

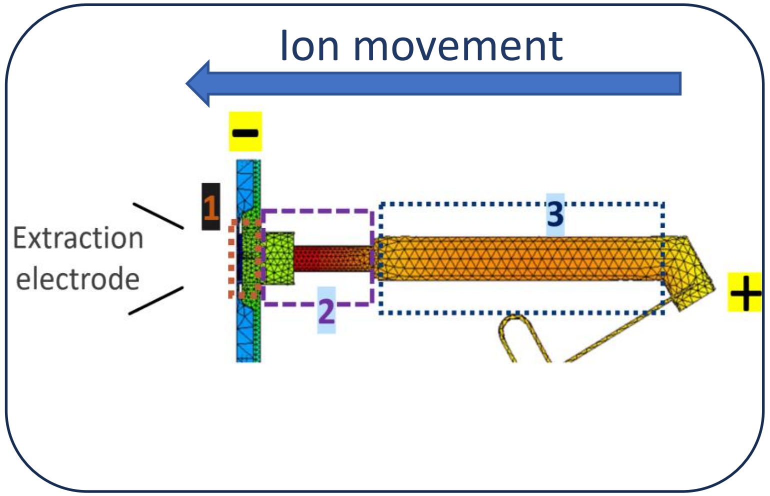
What happens if we lose all the laser ions via collision with the inner wall after their formation?

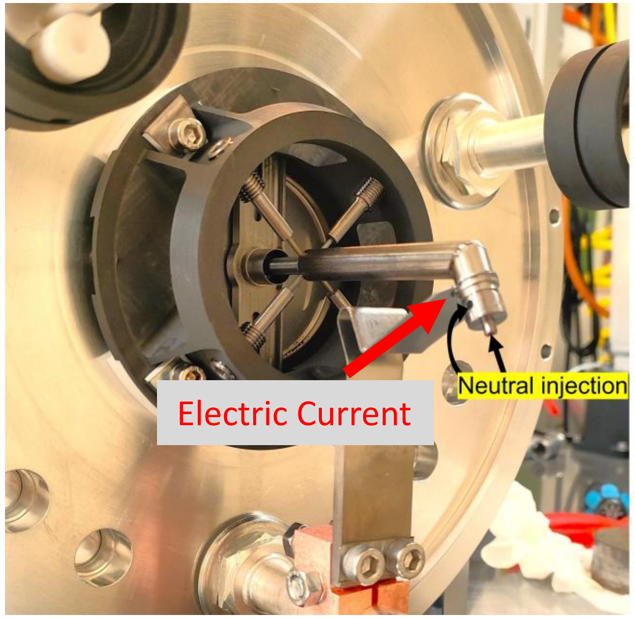
- ✓ Given the high thermal environment, this process has a high probability.
- ✓ But a **radial ion confinement** prevents this from happening, through the formation of a **quasi-neutral plasma** inside the ion source (*R. Kirchner et al.* https://doi.org/10.1016/0168-9002(90)90377-I).
- ✓ It is closely linked to the **thermionic emission from the surface wall** of the source material and hence, **high temperature** is required in such laser ion source.

This is why an ISOL Laser Ion Source is not just a Laser Ion Source but a ***"Hot Cavity" Resonant Ionization Laser Ion Source.***

Existing SPES Laser Ion Source (since 2017)

- ✓ **Material: Tantalum**
  Work function = 4.28 eV
- ✓ **Region 1: Orifice of the ion source**
- ✓ **Region 2: Hot cavity**
  Length: 33 mm
  Internal diameter: 3.1 mm
  External diameter: 5.1 mm
- ✓ **Region 3: Transfer line**
  Length: 70 mm
  Internal diameter: 8 mm
  External diameter: 8.8 mm





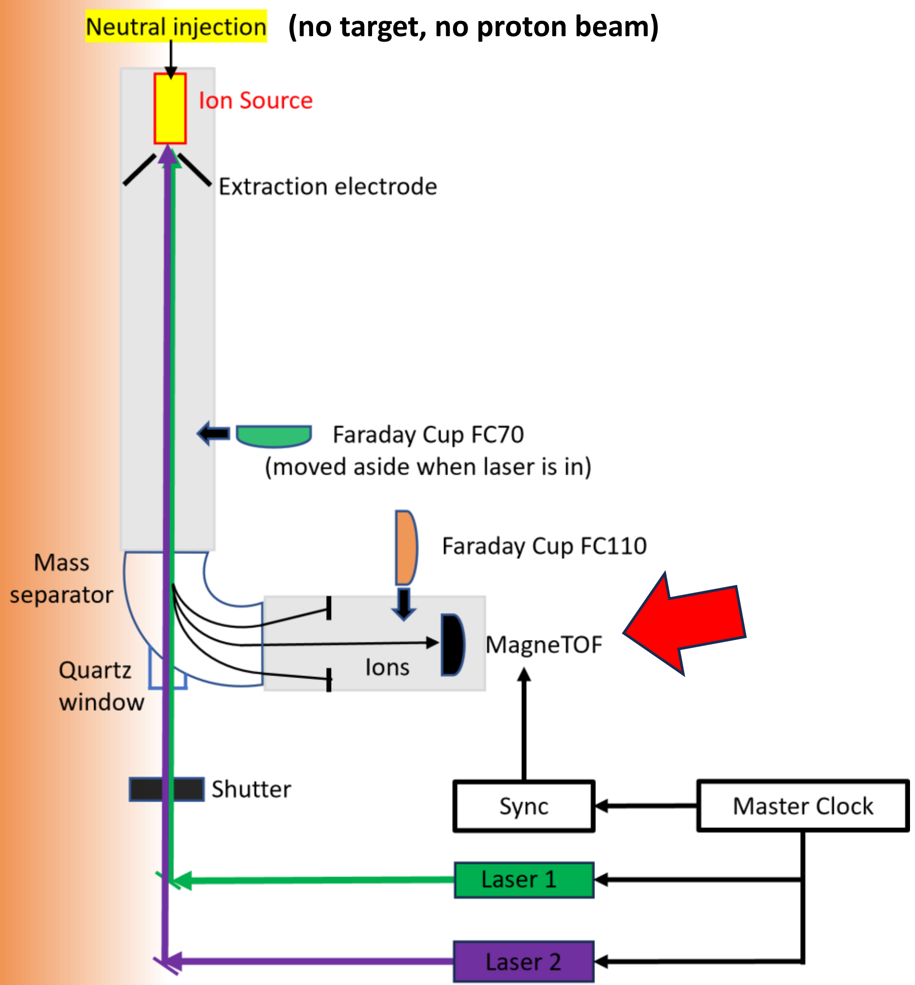
O.S. Khwairakpam et al. *https://doi.org/10.1016/j.nimb.2024.165249*

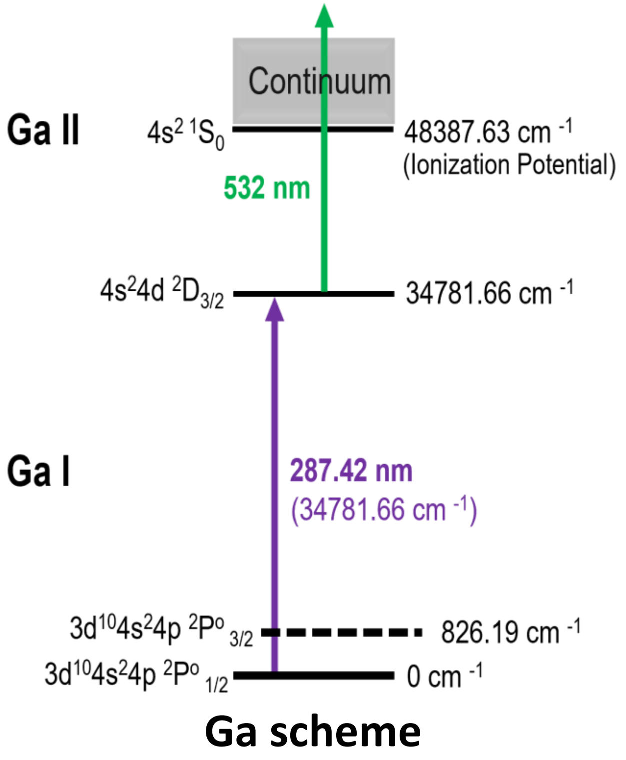
Improvements made over the previous source:

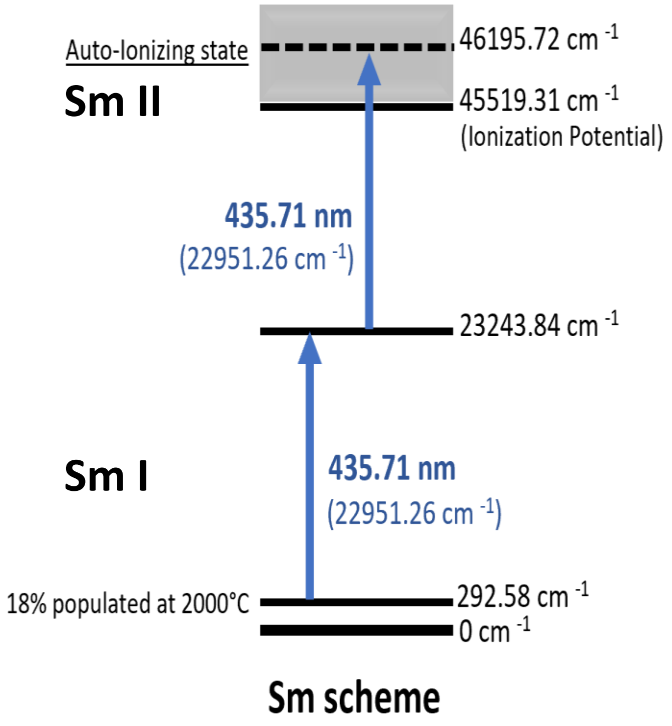
✓ Optimized thermal profile.

✓ 4 sets of tungsten centering pins to mitigate thermal deformation.

Experimental tests at ISOLDE Offline 2, CERN (2023)





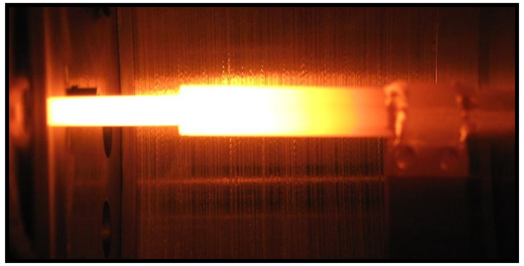


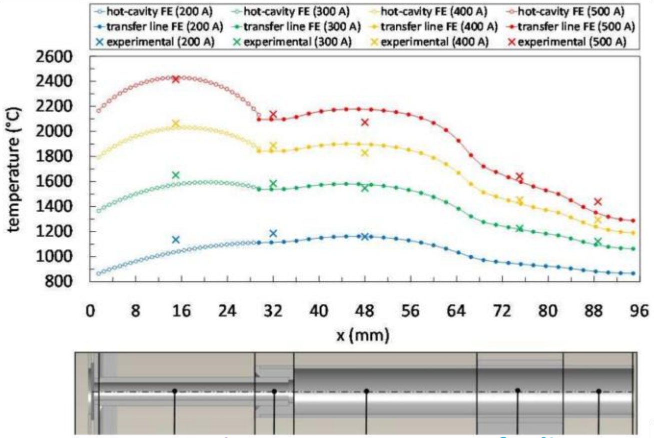
O.S. Khwairakpam et al. *https://doi.org/10.1016/j.nimb.2024.165249*

Comparison with the ISOLDE MK1 source

ISOLDE MK1 Source

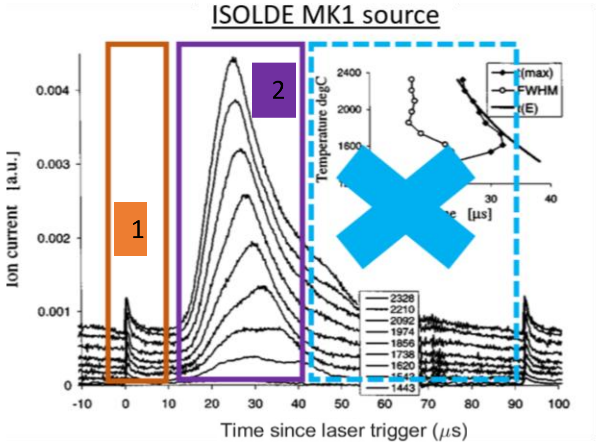
2011





a)

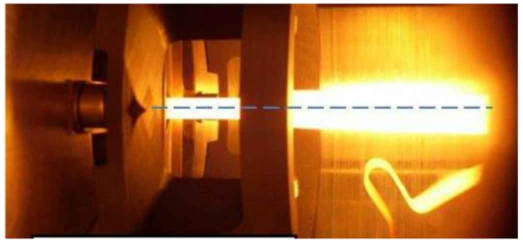
Exit (1) Hot cavity (2) Transfer line (3)



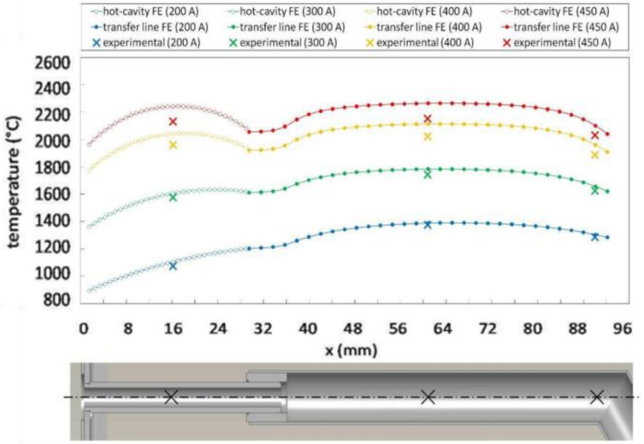
J. Lettry et al. http://dx.doi.org/10.1063/1.1148978

Existing SPES Source

2017

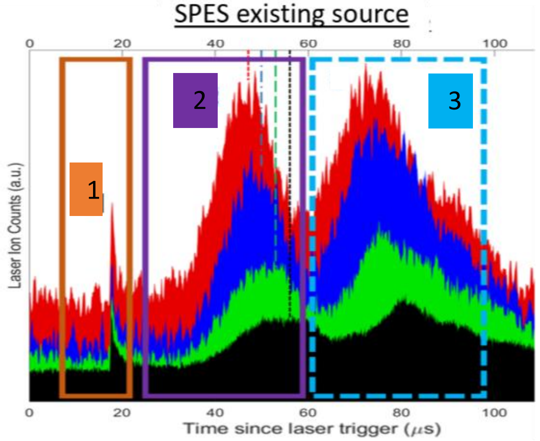


M. Manzolaro et al. http://dx.doi.org/10.1063/1.4998246



b)

Exit (1) Hot cavity (2) Transfer line (3)

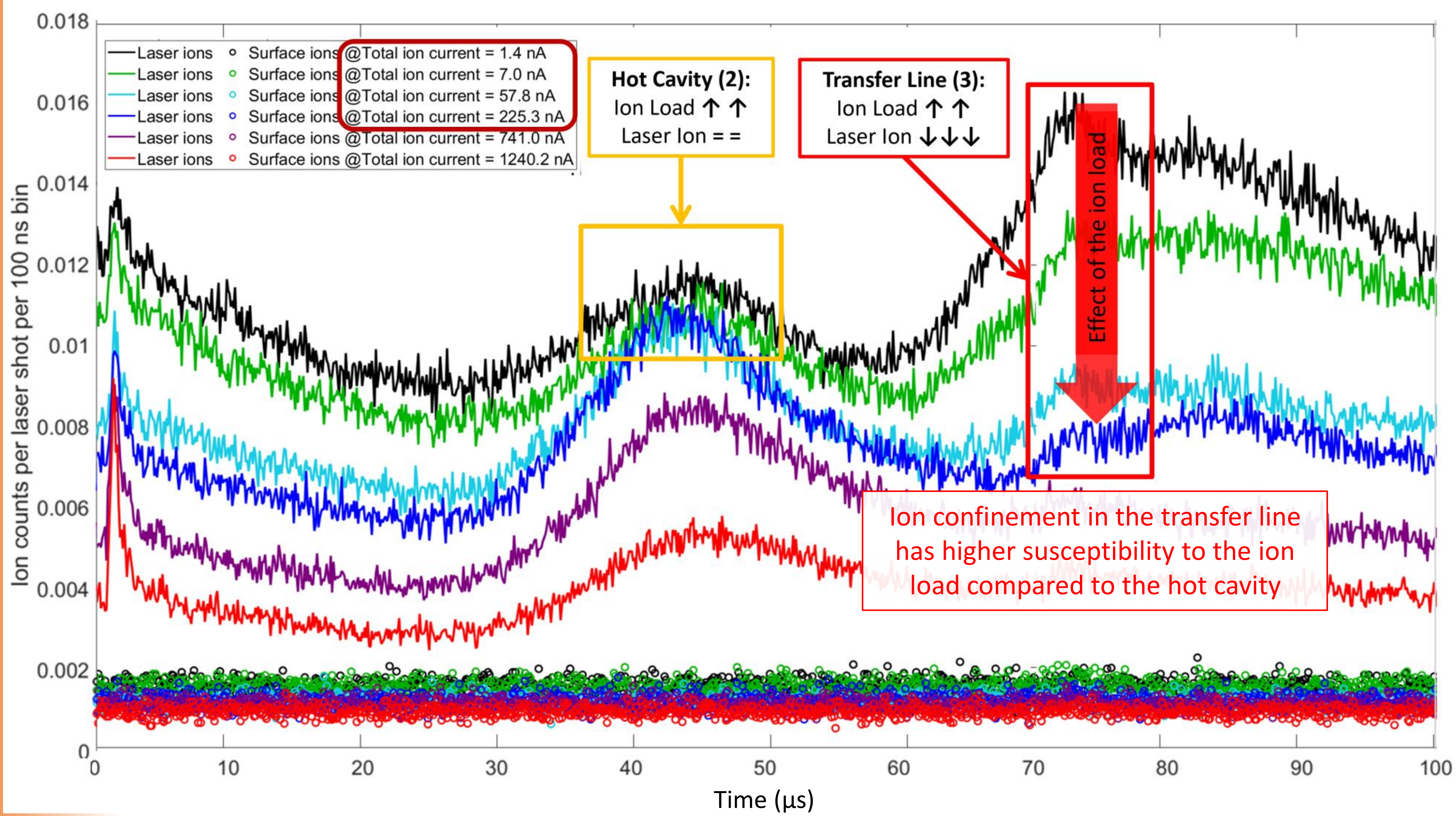


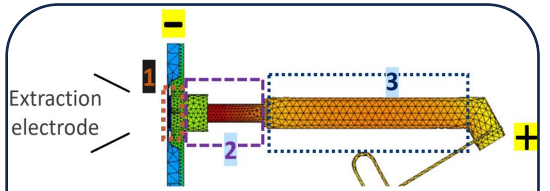
Characterized in 2023

O.S. Khwairakpam et al. https://doi.org/10.1016/j.nimb.2024.165249

What happens if we maintain the **same operational temperature**,
and **vary the level of surface ion contamination (ion load)** inside the laser ion source?

Time structure of ion beam against different ion load





1. **Orifice**
2. **Hot cavity**
3. **Transfer Line**

Ion load:
In the offline labs, we simulate the **level of contamination** present inside the ion source in real operation.

Controlled using different elements (in this case K) which are easily surface ionized.

Indicated as **Total Ion Current** (measured before the mass separation, and without lasers)

O.S. Khwairakpam et al. https://doi.org/10.1088/1742-6596/2743/1/012066

SPES Laser Ion Source: Progress Summary

v1 — 2011 → **v2** — 2017 → (characterization of v2) — 2023 → **v3** — Today

2011
- ❖ Design adapted from ISOLDE MK1 source

2017
- ❖ Thermal Optimization
- ❖ Introduction of centering pins

2023
- ❖ **Observation of ions from the transfer line** which was not the case for ISOLDE MK1 source
- ❖ However **still inferior ion survival in the transfer line** in the high ion load conditions

Today
- ❖ Optimize the transfer line for higher ion survival rate (in **high ion load conditions**)

HILLIS

Objective of the HILLIS project

Development, Construction and Characterization of a next-generation resonant laser ion source to tackle the high ion load problem.

- Deeper understanding of the **radial ion confinement** mechanism → **New geometries** of the ion source
- Installation of a time-resolved **single ion counting detector** for ion source characterization at INFN-LNL.

New geometries of the ion source

Methodology

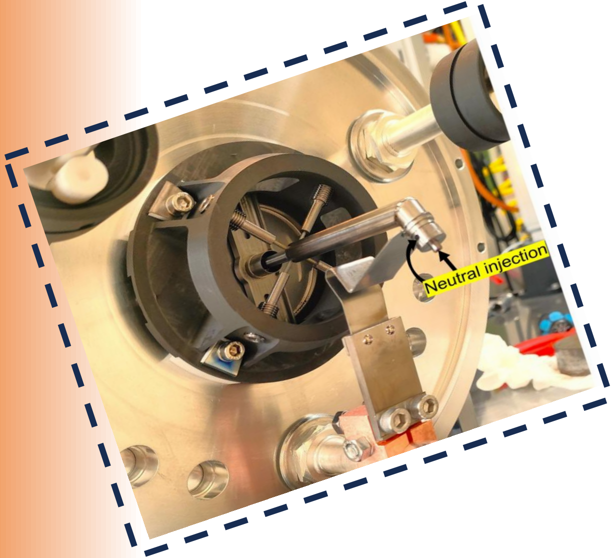
One or more new prototypes!!

Simulations
- Multiphysics Simulation
- FEM Simulation

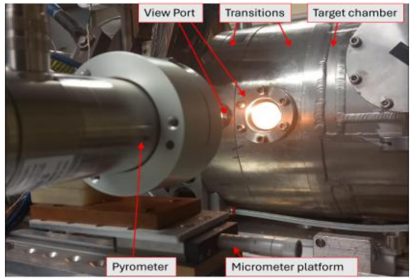
Construction
- Buy raw materials
- Mechanical construction

Characterization I
- Thermal characterization

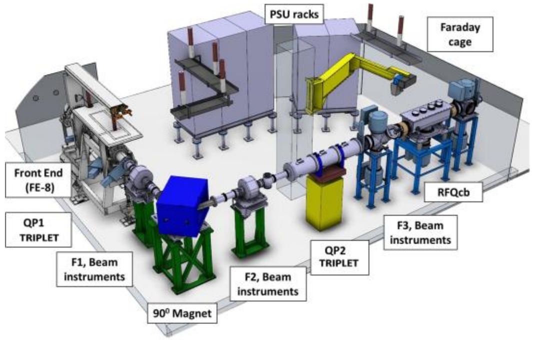
Characterization II
- Experimental tests



INFN-LNL
(in-house fabrication)



INFN-LNL



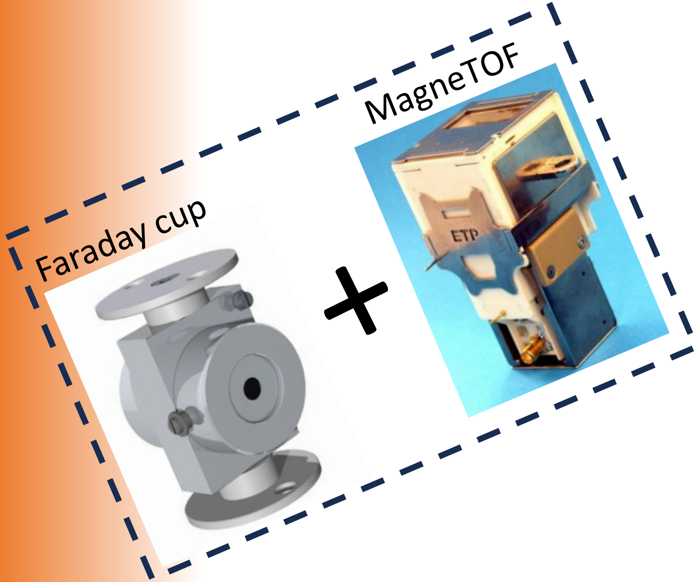
CERN

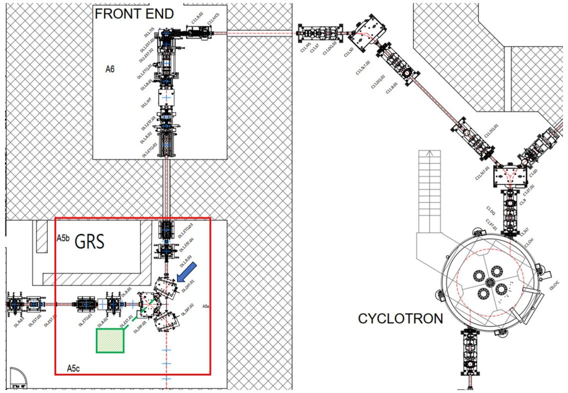
Installation of a time-resolved single ion counting detector

Methodology

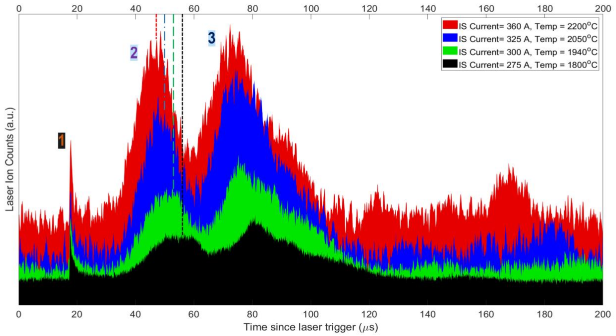
With stable beams through neutral injection into the ion source

Installation in the SPES beam line → Benchmark measurement → Future characterizations





INFN-LNL



New Ion Sources

Impact and Consequences of HILLIS

1. **ISOL community**
   - HILLIS will tackle the high ion load problems observed in ISOL ion sources.
   - Expected increase in the ionization efficiency: 50-100%.
   - This should allow lower load on the target (in terms of proton current) in certain cases.
2. **Medical communty**
   - Resonant Laser Ion Source is the standard choice as isobaric contaminants from other sources are highly risky for patients.
   - Medical radionuclides start off with an inferior in-target production, so having low efficiency is expensive.
   - MEDICIS at CERN experience very high level of ion load. Will greatly benefit from the activities foreseen in HILLIS.
3. **Ion Source community**
   - The installation of the MagneToF detector will facilitate detailed characterization of future ion sources at INFN-LNL.

Personnel and Work Packages in the HILLIS project

Name	Age	Position	Institution	FTE
Omorjit Singh Khwairakpam*	30	Post-Doc Researcher	INFN- Laboratori Nazionali di Legnaro (INFN-LNL)	0.8
Daniele Scarpa	46	Primo Tecnologo	INFN-LNL	0.5
Mattia Manzolaro	43	Primo Tecnologo	INFN-LNL	0.1
Michele Ballan	35	Tecnologo	INFN-LNL	0.1
Stefano Corradetti	40	Primo Tecnologo	INFN-LNL	0.1
Carmelo Sebastiano Gallo	39	Tecnologo		0.1
Giada Rachele Mascari	33	PhD Student	INFN-LNL	0.1
Marco Oswaldo Miglioranza	36	Tecnologo	INFN-LNL	0.05
Michele Lollo	58	Tecnico	INFN-LNL	0.05
Emilio Mariotti	64	Associate Professor	Università di Siena (Associato a INFN-LNL)	0.1
Alberto Arzenton	30	Post-Doc Researcher	Università di Padova (Associato a INFN-LNL)	0.2
Federico Pirzio	47	Associate Professor	Università di Pavia (Associato a INFN-LNL)	0.1
Total				**2.3**

* Principal Investigator

Work Packages	Activities
WP1 **Simulations and Mechanical Design of New Ion Source(s)**	- Design of new laser ion sources based on Finite Element Method (FEM) simulations of thermal and electrical profiles. - Multiphysics simulations to study radial ion confinement. - Realization of mechanical designs of the ion sources.
WP2 **Experimental tests of the New Ion Source(s)**	- Thermal and electrical characterization of the newly developed laser ion source designs. - Comprehensive experimental evaluation of resonant laser ion production performance.
WP3 **Development of Single Ion Counting Detector Set-up**	- Design and installation of the detector set-up - Benchmark measurement of the time structure of the laser ion beam

GANNT Chart and Milestones of the HILLIS project

→	Activity started	Year 1				Year 2			
●	Milestone reached	M3	M6	M9	M12	M15	M18	M21	M24
WP1	Simulations and Mechanical Design of New Ion Source(s)								
WP1.1	Multiphysics Simulation on radial ion confinement	→							●
WP1.2	FEM Simulations on the new geometry(s) of the ion source	→	●						
WP1.3	Designing and realization of the new ion source(s)		→	●					
WP1.4	Re-Optimization of the ion source using experimental feedback				→		●		
WP2	Experimental tests of the New Ion Source(s)								
WP2.1	Thermal Characterization of new transfer line at INFN-LNL			→ ●					
WP2.2	Testing of the new ion source geometry at ISOLDE Offline 2				→ ●				
WP2.3	Thermal Characterization of re-optimised ion source at INFN-LNL					→	●		
WP3	Development of Single Ion Counting Detector Set-up								
WP3.1	Design and realization of the vacuum chamber/diagnostic box	→		●					
WP3.2	Installation of MagneTOF detector in the SPES beam line			→		●			
WP3.3	Vacuum test and System Integration test of the set-up					→ ●			
WP3.4	Benchmark time structuture measurement at SPES						→ ●		
WP3.5	Testing of re-optimised ion source using the new detector set-up							→	●

No.	Milestone	Due
1	Tests of the re-designed laser ion source	12-2026
2	Installation of the single ion counting detector	03-2027
3	Tests of the ion source designs using the new detector	11-2027

No.	Deliverables	Due
1	Report on the tests of the re-designed laser ion source	02-2027
2	Report on the installation of the single ion counting detector	05-2027
3	Report on the tests of the ion source designs using the new detector	12-2027

Synergies with Other Activities

1. **HISOL_NEXT (INFN-CSN5)**
   - Connection with the ongoing research on high performance ISOL target and ion sources.
   - Ion sources: Additive manufacturing/ customized commercial pieces
2. **SPES at INFN-LNL**
   - Will provide the possibility of hosting the HILLIS detector set-up in one of the beam lines.
   - HILLIS detectors, specially the Faraday Cup can always provide a back up support in case of need.
3. **ISOLDE at CERN**
   - Has always been a strong reference point for research in this field.
   - Several on-going works on different approaches to ISOL ion source optimizations.

Costs

First year	
Missioni	4 k€
Consumabili	51.53 k€
Inventariabili	19.45 k€
Total	**74.98 k€**
Second year	
Missioni	5 k€
Consumabili	24.96 k€
Inventariabili	1.34 k€
Total	**31.30 k€**

Risk Evaluation

Adverse event	Impact	Solution/Mitigation
1. Long shut down planned for ISOLDE, CERN	Possible difficulty in finding schedule at ISOLDE Offline 2 lab	Move the experimental tests to TRIUMF, Canada >> Increase in cost of missione foreseen
2. Delay in installing the single ion counting detector due to other priorities of SPES	Difficulty in performing the task WP3.5 at SPES, INFN-LNL	Perform the planned tests with re-optimized source at ISOLDE or TRIUMF
3. Detector set-up has lower performance than expected	Difficulty in characterization of ion sources	Substitute the MagneTOF detector. >> Additional cost of 5-6k euros

Conclusion

- HILLIS project will tackle the high ion load situation, a very well-known limitation in the ISOL ion sources.
- The impact of the project will extend to international facilities such as MEDICIS at CERN.
- The project will promote in-house fabrication of ion sources at INFN-LNL.
- The project will develop a technique for comprehensive characterization of ion sources at INFN-LNL.

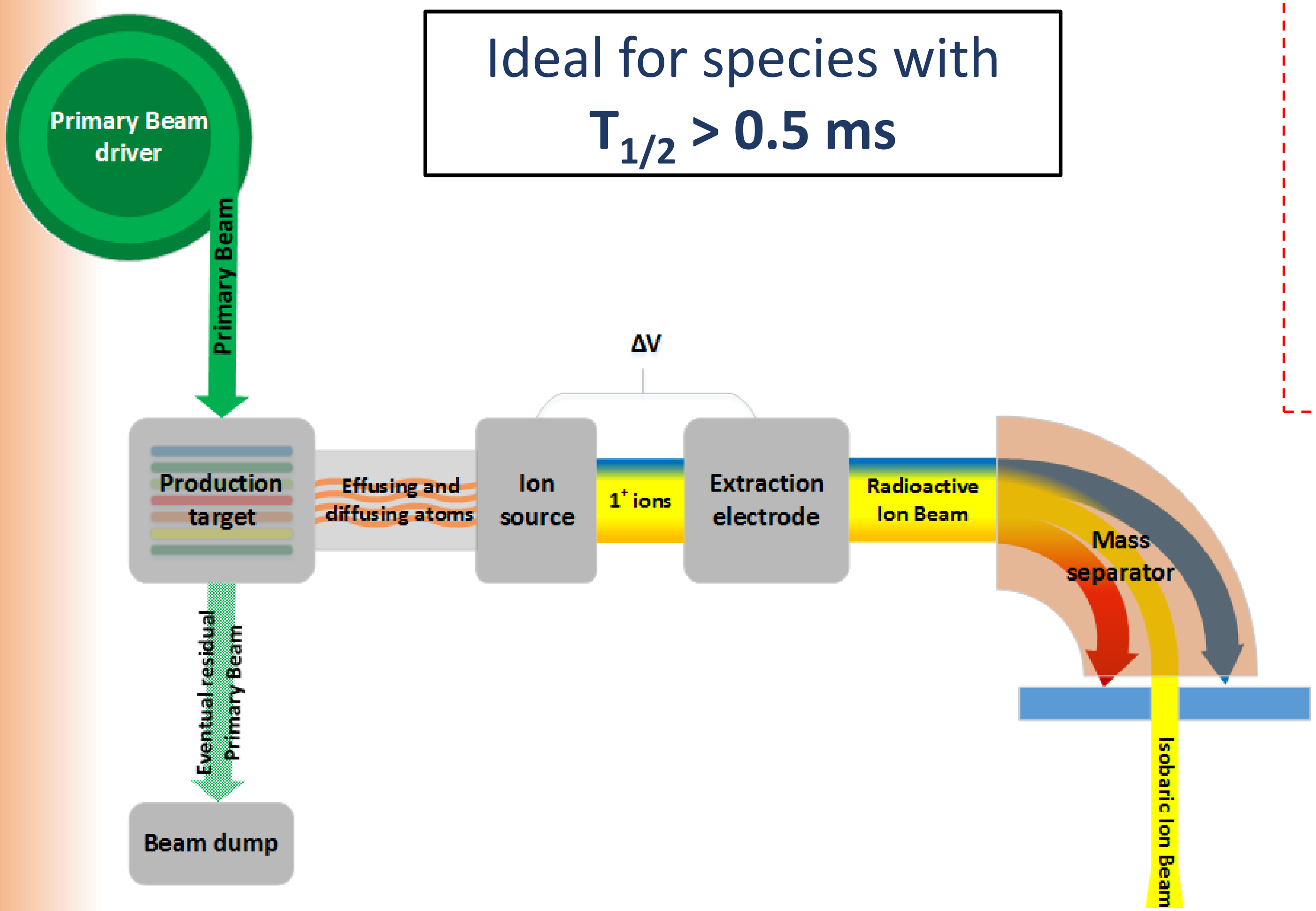

Thank You for your attention!

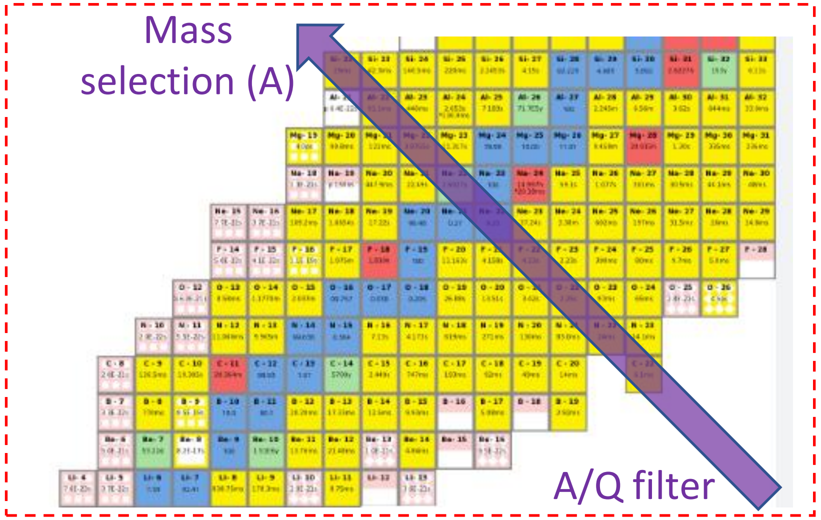
Back up slides

A simplified layout of an ISOL facility

Ideal for species with

$T_{1/2} > 0.5$ ms





Isobaric contamination situation

For the UC_x target with proton energy 40 MeV

	^{87}Ga	^{87}As	^{87}Se	^{87}Br	^{87}Kr	^{87}Rb	^{87}Sr
Type:	**Desired**	Not Desired	Not Desired	Not Desired	Not Desired	Not Desired	Not Desired
$T_{1/2}$:	95 ms	484 ms	5.5 s	55.68 s	76.3 s	**Stable**	**Stable**
In-target production ratio:	**1**	**~10^3**	**~10^3**	**~10^4**	**~10^4**	**~10^3**	**~10^2**

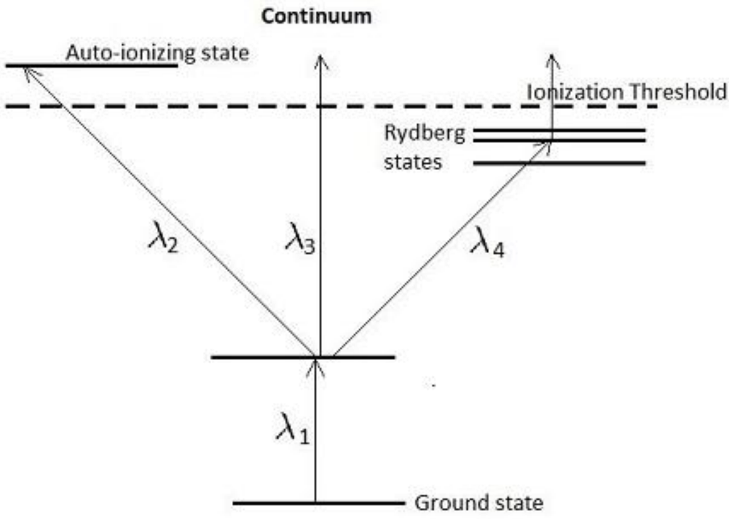
>> For the particular case of ^{87}Ga, the level of isobaric contamination is high as shown above.
And it is the case for many other isotopes.

Calls for an **element selective ionization technique** to attain high-purity isotopic beams → **Resonant Ionization Laser Ion source (RILIS)**

What separates the Laser Ion Source from other sources?

>>Resonant technique of ionization<<

- ✓ Surface and plasma ion sources looks essentially at the **ionization threshold** as one unit/limit that needs to be crossed at one step.
- ✓ Laser ion source breaks down this process **into several steps** combined. Each step is unique to the electronic configuration of the element considered.
- ✓ Thus, so called the "**Resonant Ionization Laser Ion Source (RILIS)**".



D. Scarpa et al. *https://doi.org/10.1063/5.0078913*

Characterization using the Time Structure of the ion beam

- ✓ **Time structure** of an ion beam is measured using **single ion counting** method.
- ✓ The ion count is performed inside a time window **synchronized with the laser pulses @5 kHz or 10 kHz.**
- ✓ Basically, putting **a time-stamp on each ion** arriving at the MagneTOF detector with reference to the pulses of the laser and making **a histogram over thousands of laser shots**.
- ✓ It provides information regarding the ion source environment.

Inside the Resonant Ionization Laser Ion Source (RILIS)

RILIS

Laser

Laser lab at floor +1

Hot Cavity

Front End at floor -1

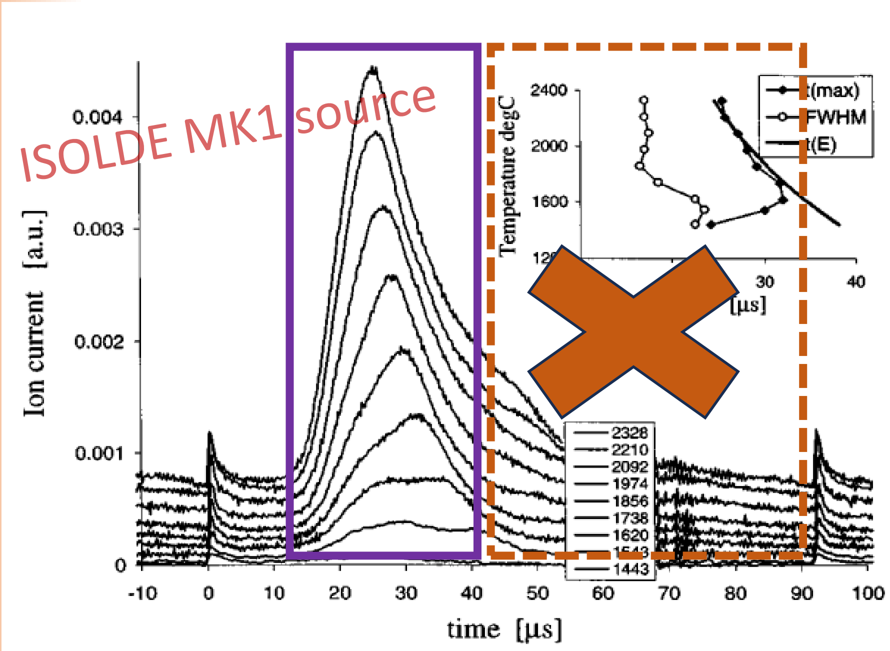
~ 20 m

- ✓ Atomic transitions
- ✓ Multi-step ionization scheme development
- ✓ Laser wavelength
- ✓ Laser linewidth
- ✓ Laser power

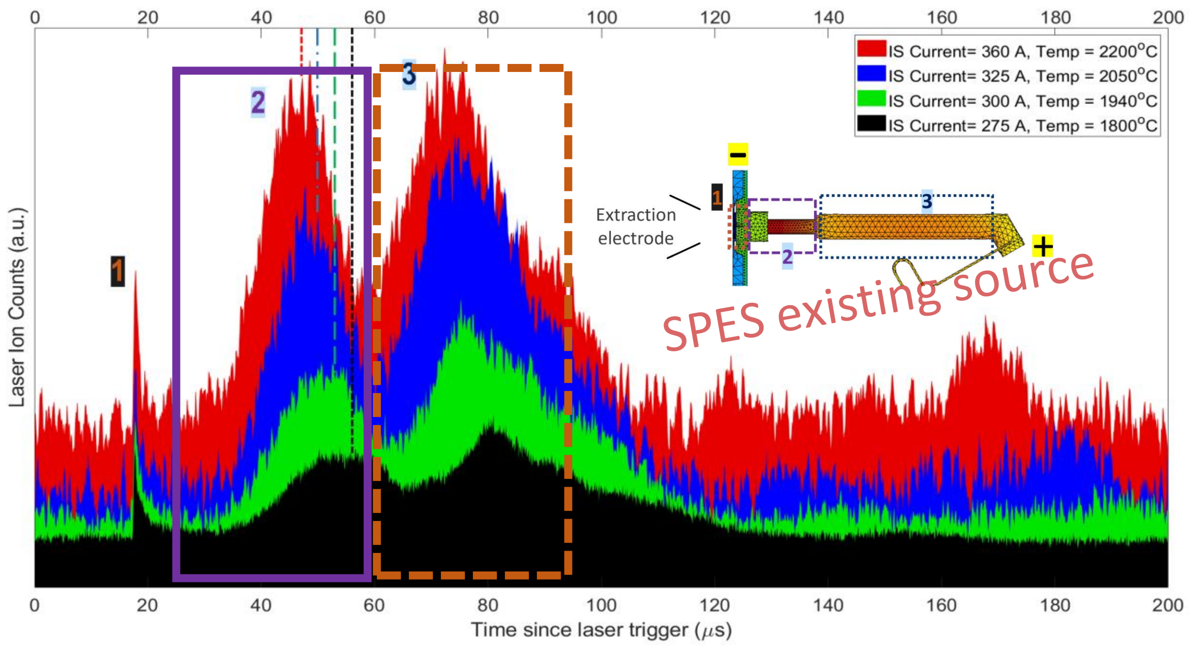
- ✓ Beam transport
- ✓ Stability of beam position
- ✓ Mitigation of vibrational noise

- ✓ Geometry
- ✓ Cavity material
- ✓ Thermal behavior
- ✓ Ion survival rate

Comparison with the ISOLDE source



J. Lettry et al. http://dx.doi.org/10.1063/1.1148978



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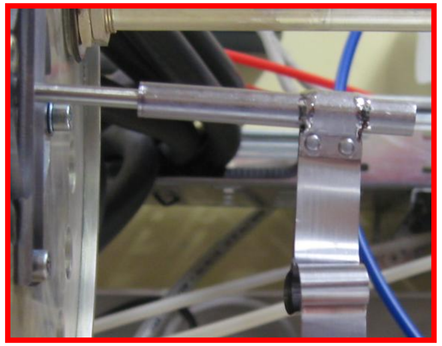
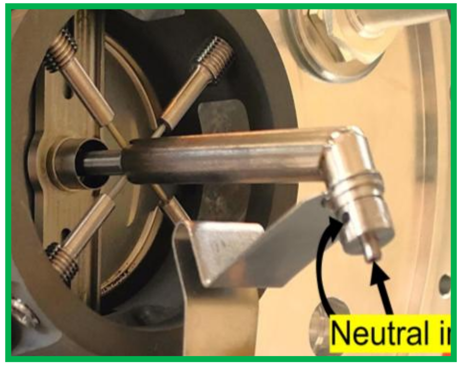


Why?

- ❌ Axial electric field in the whole length ✓
- ❌ High temperature in the transfer line ✓



How to approach?

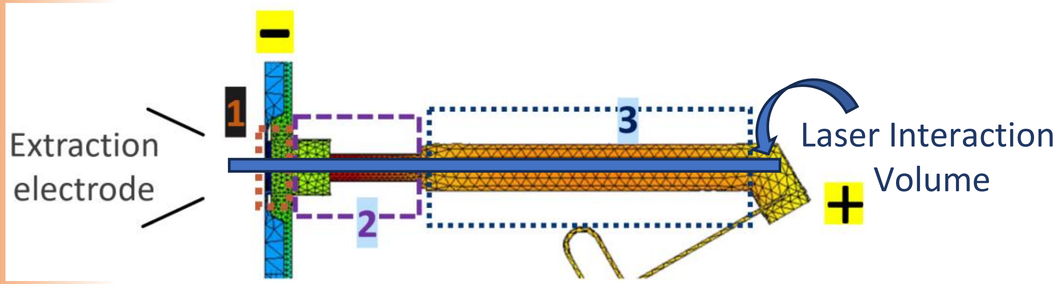
We want the radial ion confinement to resist the high ion load effect in the transfer line.

Parameter	Hot cavity (2)	Transfer Line (3)	Discrepancy
Temperature (°C)	⚓ 2200	2000 ⚓	-10%
Diameter (mm)	⚓ 3.1	8	+150%

The approach is to understand and optimize **the effect of the geometry (diameter)** on the radial ion confinement.

✓ Keep temperature the same and **change just the geometry of the transfer line.**

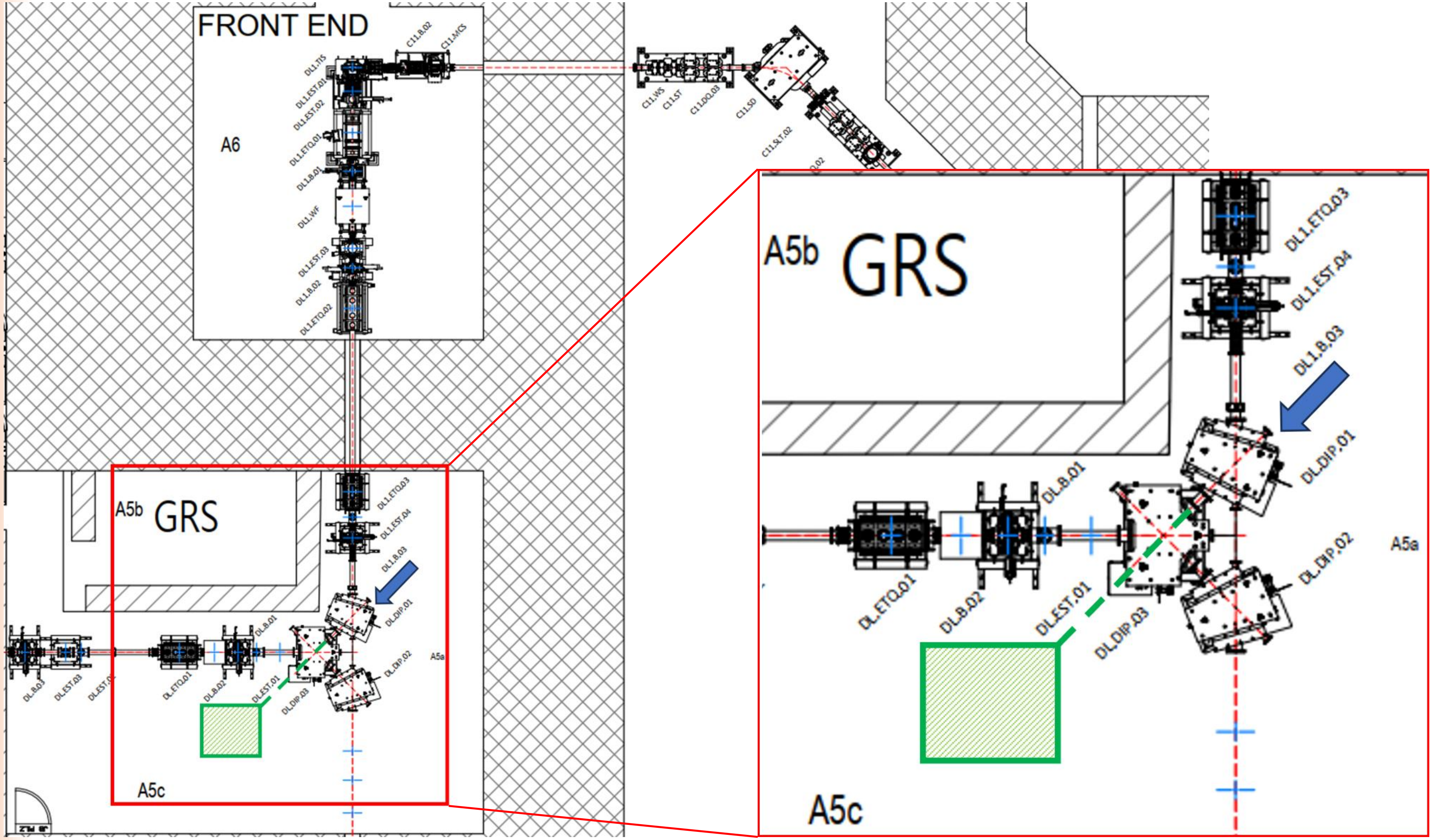
✓ This doesn't compromise the existing laser interaction volume. Indeed, we have **roughly 80% of unused volume in the source**.



Follow-up question: Why not make a single tube with the dimension of hot cavity?

A bit of extra "room" in the transfer line allows to maintain a similar laser interaction volume, in the case of thermal deformation of the line.

Installation of a time-resolved single ion counting detector



Istituto Nazionale di Fisica Nucleare
LABORATORI NAZIONALI DI LEGNARO

O.S. (Omi) Khwairakpam

Università degli Studi di Padova

New geometries of the ion source

Materials / Travels

Ion Source Materials

1. Tantalum rod (hot cavity)
2. Tantalum tube (transfer line)
3. Tantalum capillaries (mass marker)
4. Mass marker solution

Travels

Experimental tests at ISOLDE Offline 2, CERN

Installation of a time-resolved single ion counting detector

Materials

Detector part

1. MagneTOF (single ion counting, res < 1ns)
2. TimeTagger4 TDC module (res < 1ns)
3. Faraday Cup (high sensitivity)
4. Picoammeter (20 fA to 20 mA)
5. Computer

Vacuum part

1. Vacuum chamber
2. Vacuum gauges
3. Turbo pump
4. Gate valve